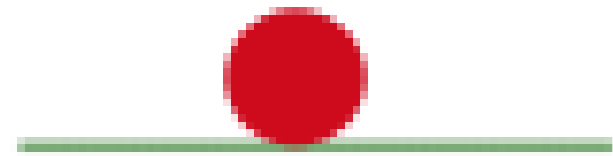
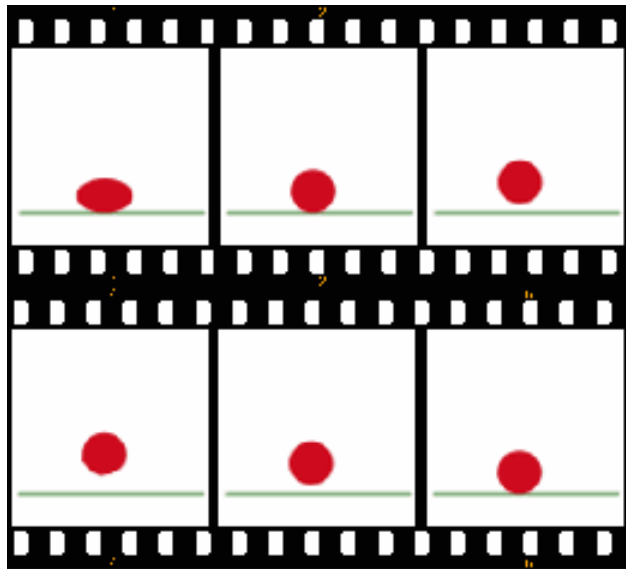




# Introduction to animation

BY: DR SITI HAZYANTI

# WHAT IS ANIMATION?

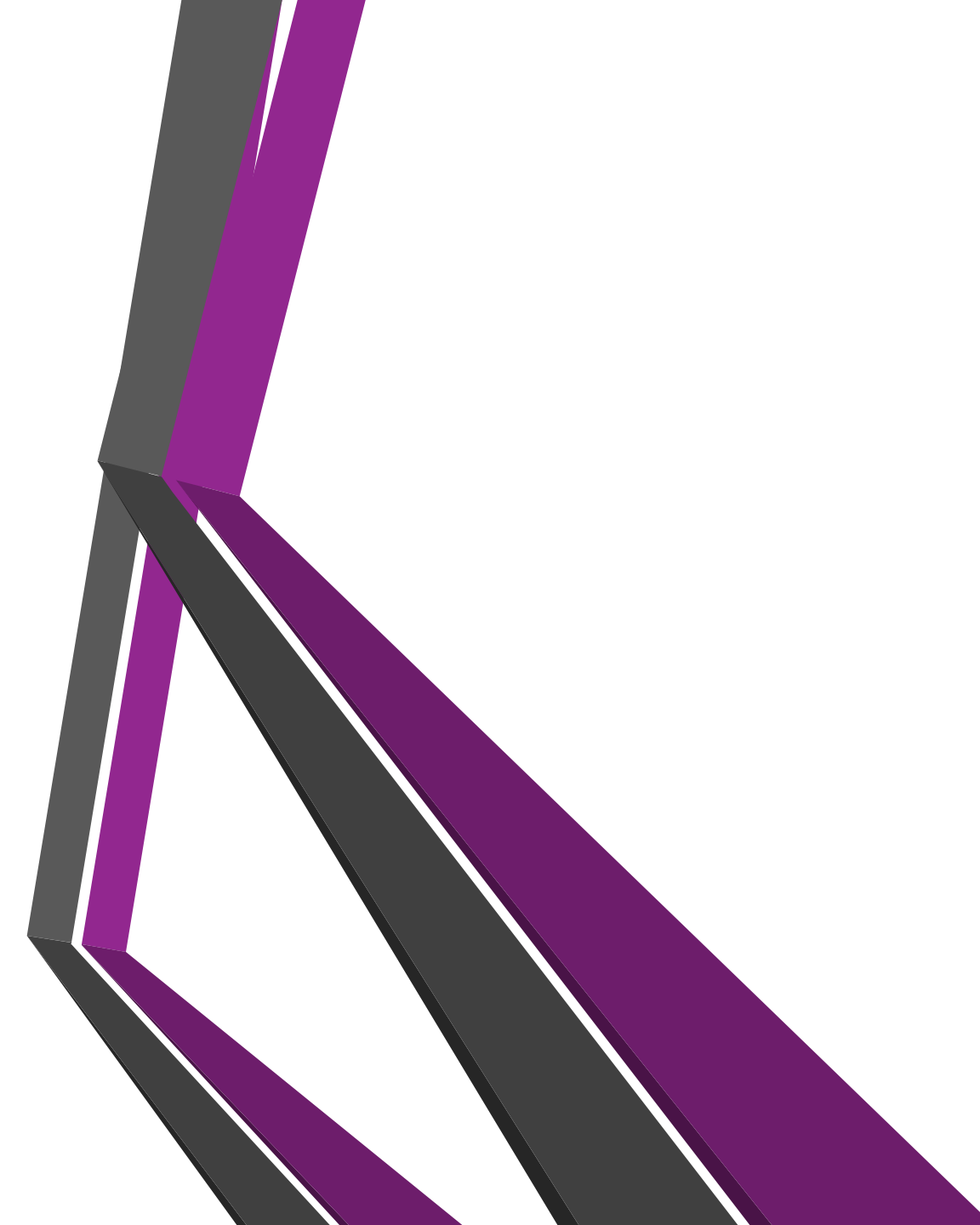


# WHAT IS ANIMATION?

- **Animation** is the rapid display of a sequence of images of 2-D or 3-D artwork in order to create an illusion of movement.
- Animation is the art of movement expressed with images that are not taken directly from reality
- It is an optical illusion of motion due to the phenomenon of persistence of vision.

# WHAT IS ANIMATION

- To animate means to give life to an inanimate object, image, or drawing
- In animation, the illusion of movement is achieved by rapidly displaying many still images or frames in sequence.
- The most common method of presenting animation is as a motion picture or video program.



# 3D COMPUTER ANIMATION

# WHAT IS 3D COMPUTER ANIMATION

- Computer animation consists of a series of individual images.
- These images are stored one at a time in the computer. They are viewed at the normal playback speed of 30 frames per second. The result is a moving picture, or animation.
- [3D ANIMATION](#)

# WHAT IS 3D COMPUTER ANIMATION

- 3-D Computer Animation is the world's fastest growing creative technology.
- 3-D dominates in the animated feature, game, special effect, interactive, and commercial industry
- Computer 3-D animation is taking over the film world
  - Computer Animated are breaking box office records
  - Computers are used for many of the special effects in live-action films, and digitized actors.
- The field of 3-D animated video, computer, and interactive games has grown amazingly

An abstract geometric design in the bottom-left corner of the slide. It consists of several overlapping, elongated triangular and quadrilateral shapes. The primary colors are a vibrant purple and a dark charcoal grey. Thin white lines are used as outlines or dividers between some of the overlapping shapes, creating a sense of depth and complexity. The design is modern and angular, typical of contemporary graphic design.

# CAREERS IN COMPUTER ANIMATION

# CAREERS in Computer Animation

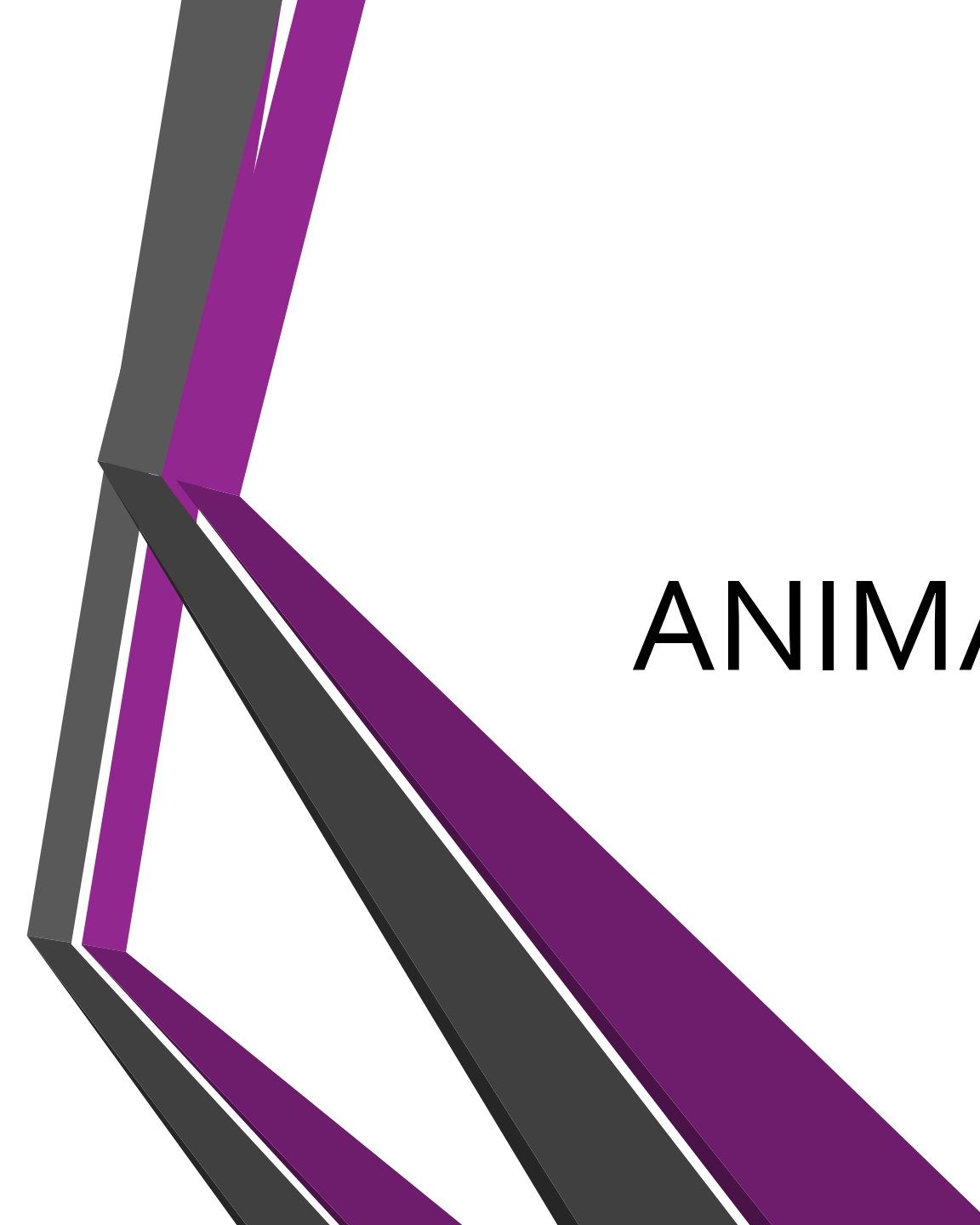
- 3D Computer Modeler
- Animator
- Art Director
- Graphic Artist
- Interface Designer
- Multi Media Design Engineer

# Benefits Provided by 3-D Computer Animation

1. A computer animation can graphically simplify complex concepts which are difficult to visualize.
2. 3-D computer animation captures attention.
3. It can communicate more information than an oral description can.
4. information presented as moving images is retained by the viewer for a longer time and with greater accuracy

# Benefits Provided by 3-D Computer Animation

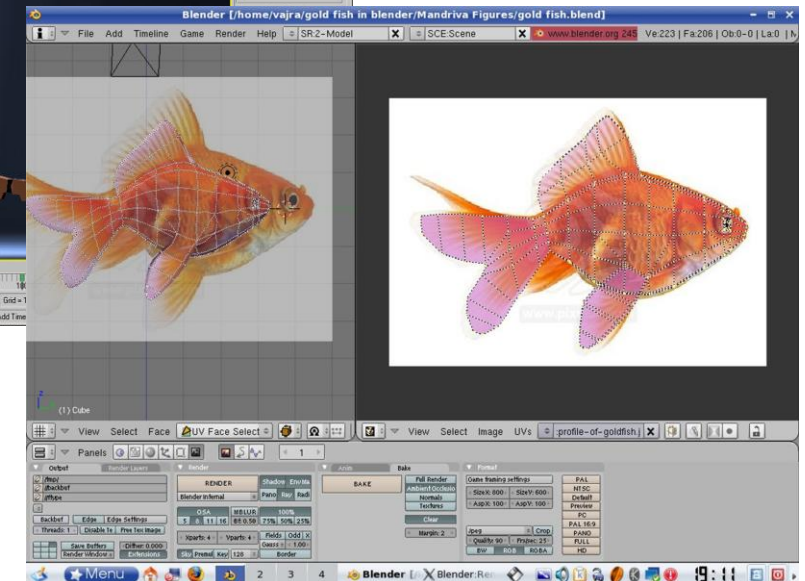
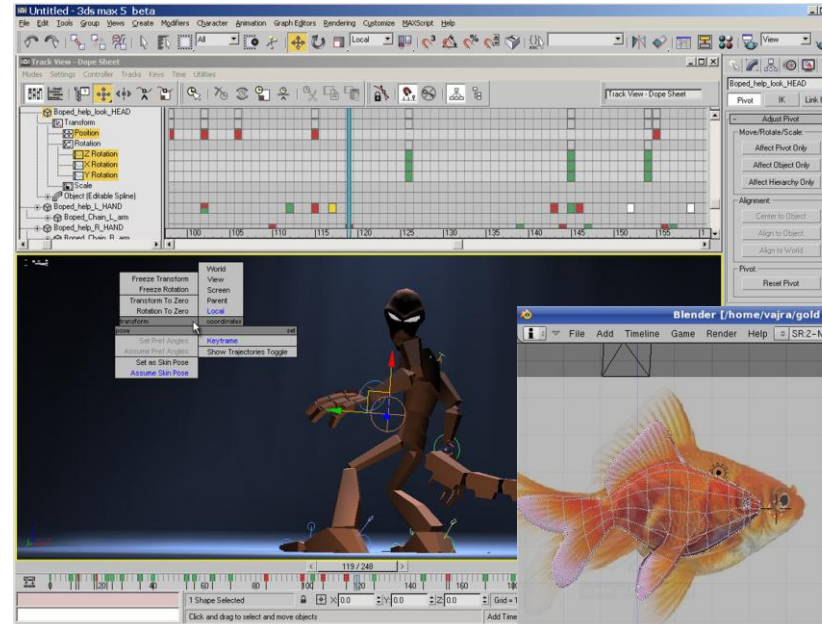
5. A computer animation can re-create an event which is too expensive or too dangerous to reproduce, such as an aircraft accident.
6. It can re-create a scene which has been altered or which no longer exists, such as a building which has been demolished.
7. When used as an illustrative tool, computer animation can help the presenter maintain focus. It creates interesting presentation.



# 3D COMPUTER ANIMATION SOFTWARE

# 3D Computer Animation Software

1. **Blender**
2. **Maya**
3. **3DS MAX**
4. **Softimage**
5. **Lightwave**
6. **Animation Studio**



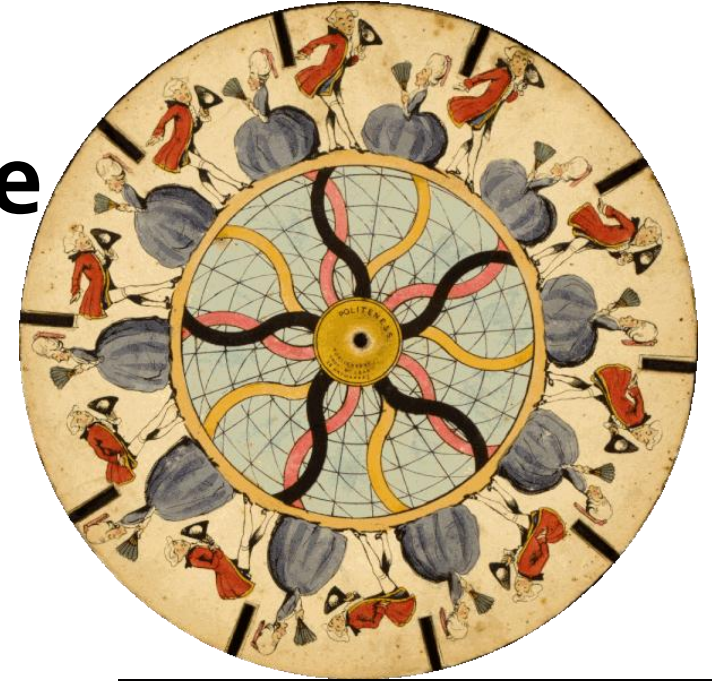
An abstract geometric design in the bottom-left corner of the slide. It consists of several overlapping, elongated triangular and quadrilateral shapes in shades of purple and grey, creating a sense of depth and movement.

# SHORT HISTORY OF ANIMATION

# 1. Phenakistoscope

- **1872** Phenakistoscope invented  
- produced an illusion of movement

- <https://www.vectornator.io/blog/phenakistoscope/>



## 2. Funny Faces

- 1906 J. Stuart Blackton made the first animated film which he called "*Humorous phases of funny faces.*"
- His method was to draw comical faces on a blackboard and film them. He would stop the film, erase one face to draw another, and then film the newly drawn face.
- [https://www.google.com/search?q=humorous+phases+of+funny+faces&source=lnms&tbm=vid&sa=X&ved=2ahUKEwiSqZGX5P7gAhURVWwGHdWiDRAQ\\_AUoAnoECAEQBA&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:a62d7868,vid:wGh6maN4I2I](https://www.google.com/search?q=humorous+phases+of+funny+faces&source=lnms&tbm=vid&sa=X&ved=2ahUKEwiSqZGX5P7gAhURVWwGHdWiDRAQ_AUoAnoECAEQBA&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:a62d7868,vid:wGh6maN4I2I)

### 3. Little Nemo

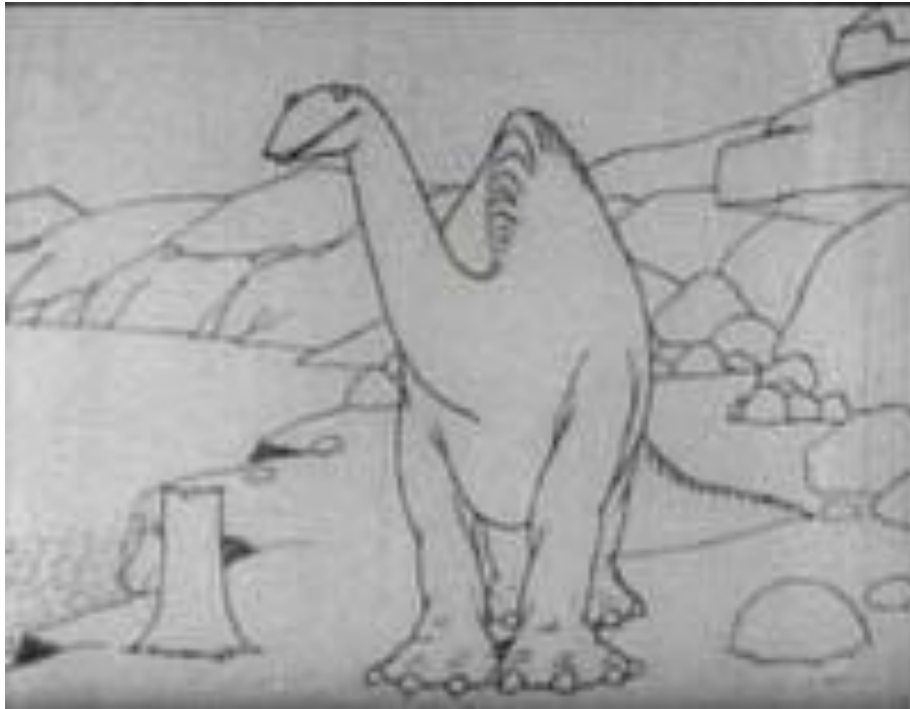


- 1911 Winsor McCay produced an animation sequence using his comic strip character "Little Nemo."

- It was colorized by hand

- [https://www.imdb.com/video/vi3717595417/?playlistId=tt0104740&ref=tt\\_ov\\_vi](https://www.imdb.com/video/vi3717595417/?playlistId=tt0104740&ref=tt_ov_vi)

## 4. Gertie, The Trained Dinosaur



[https://www.youtube.com/watch?v=32pzHWUTcPc&ab\\_channel=OpenCulture](https://www.youtube.com/watch?v=32pzHWUTcPc&ab_channel=OpenCulture)

- Winsor McCay produced a cartoon called "Gertie, The Trained Dinosaur" which amazingly consisted of 10,000 drawings.
- It was shown as a film in the theaters and also as a multi media event on stage with McCay interacting with the animated Gertie.

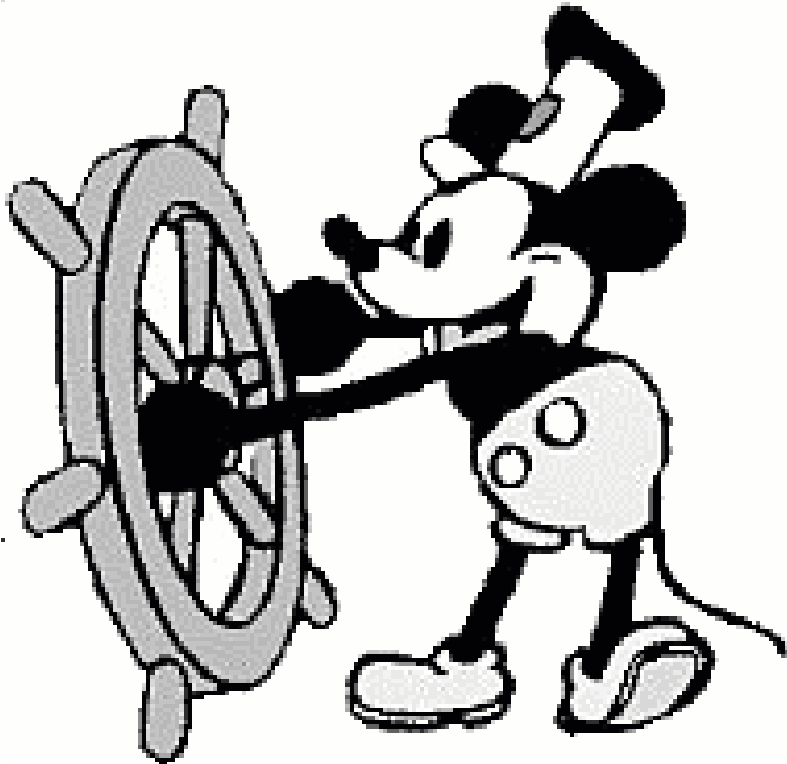
## 5. Felix the Cat



Copyright © Felix the Cat Creations Inc.

- 1920 FELIX THE CAT, the most popular character and series of this period, started as the Feline Follies from Sullivan's studio.
- [https://www.google.com/search?q=felix+the+cat&source=lnms&tbm=vid&sa=X&ved=2ahUKEwilsL45f79AhVHcWwGHTsdB6IQ\\_AUoAnoECAIQBA&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:3ed6oa49,vid:amGbBFsiuzc](https://www.google.com/search?q=felix+the+cat&source=lnms&tbm=vid&sa=X&ved=2ahUKEwilsL45f79AhVHcWwGHTsdB6IQ_AUoAnoECAIQBA&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:3ed6oa49,vid:amGbBFsiuzc)

## 6. Mickey Mouse



**STEAMBOAT WILLIE**

DR. SITI HAZYANTI

- 1928 STEAMBOAT WILLIE was the first successful sound animated film; it made Mickey an international star, and launched the Disney studio of today.
- When Disney used sound for the first time in Steamboat Willie (1928), he gained an advantage over his competitors.

- According to Disney, "Mickey Mouse's official birthday is November 18, 1928 when he made his first film debut in Steamboat Willie. This was the first Mickey Mouse cartoon released.
- Mickey Mouse's image is the most reproduced in the world.

<https://www.google.com/search?q=steamboat+willie&source=lnms&tbm=vid&sa=X&ved=2ahUKEwiLuPeZ5v79AhW1TmwGHadZBpYQAUoAnoECAEQBA&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:94110720,vid:BBgghnQF6E4>



DR. SITI HAZYANTI

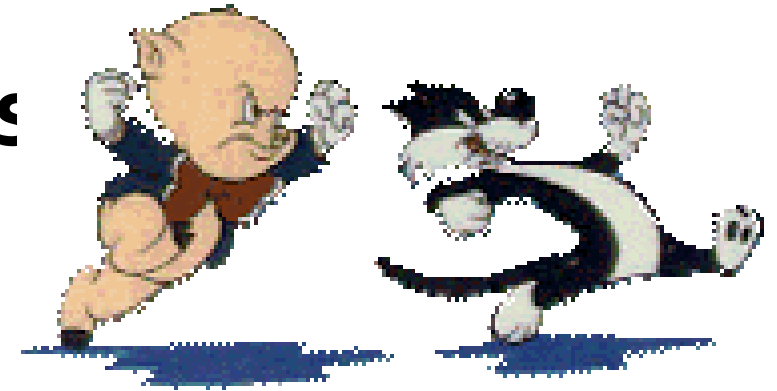
## 7. Betty Boop

In 1930 Betty Boop was first created as a dog character by Grim Natwick and appeared as the girl friend of another dog named Bimbo.

- Betty's figure was modeled after Mae West's and her singing style taken from Helen Kane the "Boop-Boop-a-Doop" girl.
- [https://www.google.com/search?q=Boop-Boop-a-Doop%22&source=Inms&tbn=vid&sa=X&ved=2ahUKEwidiqba5v79AhWjSWwGHanLBdgO\\_AUoAXoECAEQAw&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:e74cof8e,vid:9oowLfotkjE](https://www.google.com/search?q=Boop-Boop-a-Doop%22&source=Inms&tbn=vid&sa=X&ved=2ahUKEwidiqba5v79AhWjSWwGHanLBdgO_AUoAXoECAEQAw&biw=1536&bih=688&dpr=1.25#fpstate=ive&vld=cid:e74cof8e,vid:9oowLfotkjE)



## 8. Looney Toons

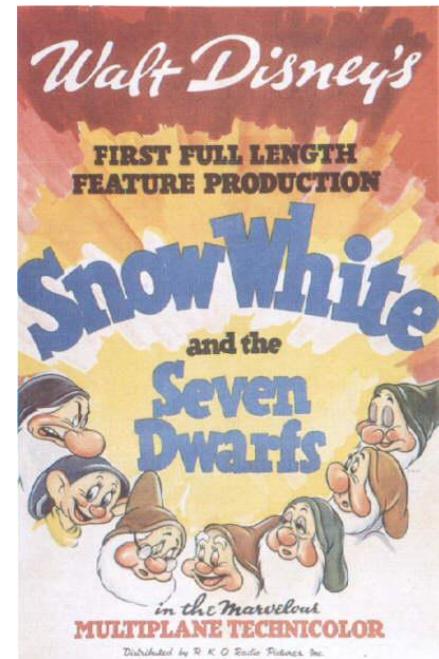


- The Looney Tunes series, created by Hugh Harman and Rudolph Ising, was introduced in 1930. A blatant rip-off of Disney's Silly Symphonies series, each Looney Tune was required to have one full chorus from a song from a Warner feature film.

## 9. Snow White

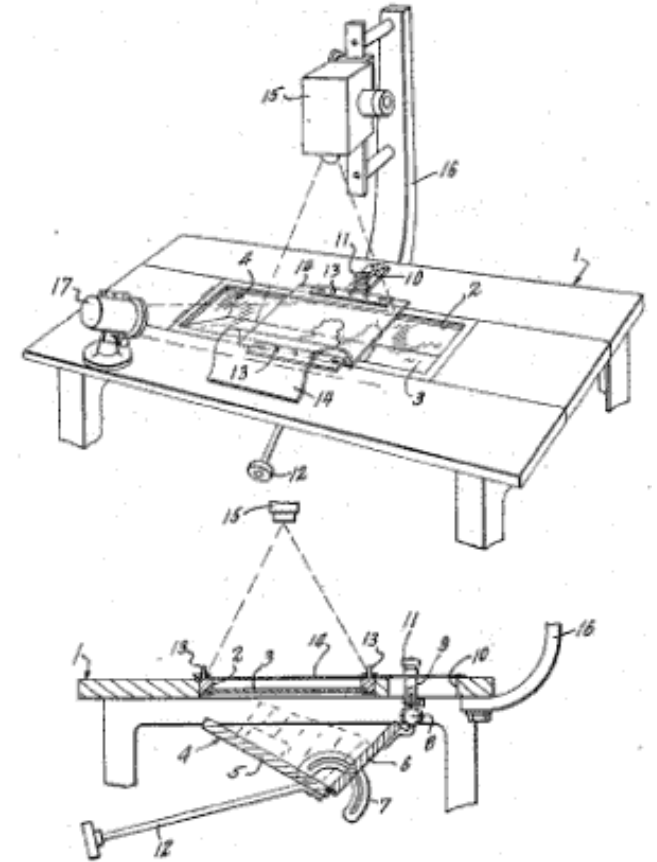
- 1937 SNOW WHITE AND THE SEVEN DWARFS

DR. SITI HAZYANTI



## 10. Walt Disney

- Walt Disney invented both Mickey Mouse and the multiplane camera. Disney was granted U.S patent 2,201,689 for the "art on animation" camera on May 1, 1940.



# King Kong

- Willis O'Brian of *King Kong* fame is usually considered the dean of stop-action animation.



# Stop Motion Animation

- Typically stop motion animation is being used to animate things that are not life-size. Some things don't scale down well or stay under your control, eg water, so substitutions need to be made.

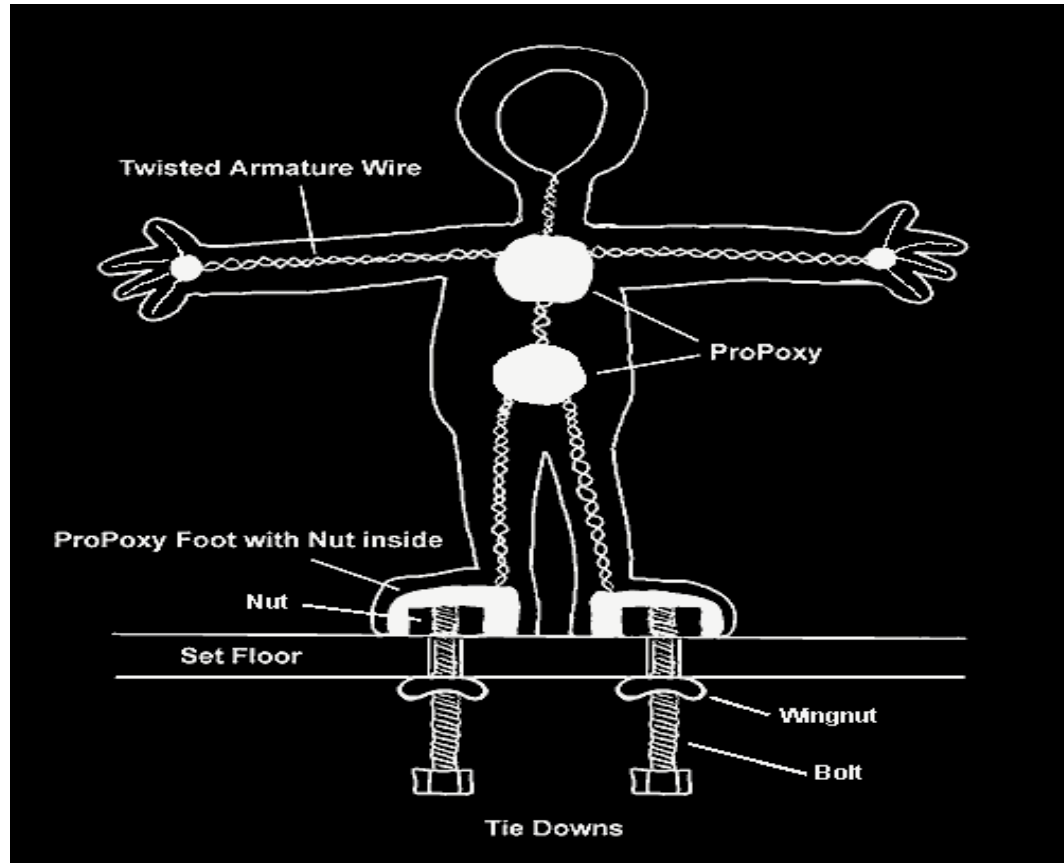


# Stop Motion Animation



- Stop-motion animation was frequently combined with live action scenes.

# Stop Motion Animation



- Machines were originally done by hand as well, but in the 70s computer controlled motion rigs took over this job. Here the computers were programmed to move the model in a given, repeatable, pattern.

- Example:

[https://www.youtube.com/watch?v=XozioUYugMA&ab\\_channel=BrainstormProductions](https://www.youtube.com/watch?v=XozioUYugMA&ab_channel=BrainstormProductions)

# Claymation

Claymation is the art of making clay figures move, talk, sing, dance or whatever your imagination wants it to do!



# Claymation

- Sometimes called plasticine, figures and props are molded from clay and are then used to tell a story. Using stop motion photography the frames are then run together to produce an animation.

# Computer Animation



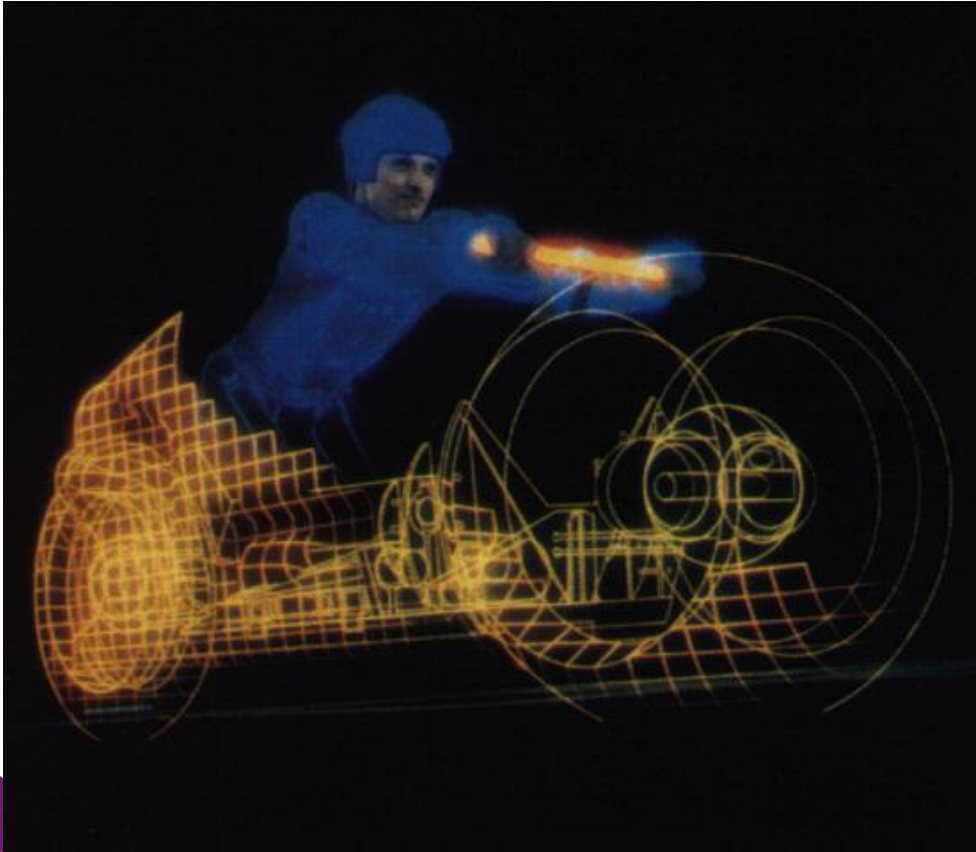
- Computer animation is considered by many to be actually closer to other animation techniques rather than traditional hand-drawn animation.
- Often it is compared to stop motion animation

# Computer Graphics

- Early on, Computer Graphics (CG) appeared in a variety of movies in which it was used as computer graphics. For example, *Future World* (1976) and *Star Wars* (1977, Image West) fall into this category.



# Tron



- *Tron* (1982) integrated computer animation with live action, but, since the action took place in a computer, the CG didn't have to look realistic .This was the first time CG was used as an integral part of a movie.

# Jurassic Park

- *Jurassic Park* (1993) was the first to completely integrate use of CG character animation in which the graphics were designed so as to blend in with the live action so that it was difficult to tell what was computer generated and what wasn't.



# Computer Animation Animation



- Of course, one use of computer animation is simply to 'do animation.' By that it mean computer animation is used to produce animated pieces which would otherwise be done by more traditional means - essentially 3D cartoons

# Pixar

- **Toy Story** (1995, Pixar, Disney), the first full length fully computer-generated 3D animation, would fall into a category of 3D cartoon



# Storyboards

- Disney's other innovations in animation technology were the use of a storyboard to review the story and the use of pencil sketches to review motion.

